

\* Rank 0 tensor.

$$[Q]$$

$$Q = Q_s^{(1)}$$

\* Rank 1 tensor. \* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = -Q_{du}^{(1)} + Q_s^{(1)}$$

$$Q_{zz} = 2Q_{du}^{(1)} + Q_s^{(1)}$$

\* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & Q_{xy} & 0 \\ -Q_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xy} = G_{pz}^{(1)}$$

\* Rank 3 tensor (ss). \* Rank 3 tensor (sa). \* Rank 3 tensor (a). \* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & 0 \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ Q_{xxyz} & -Q_{xxyz} & 0 & Q_{yzyz} & 0 & -Q_{xxzx} \\ Q_{xxzx} & -Q_{xxzx} & 0 & 0 & Q_{yzyz} & Q_{xxyz} \\ 0 & 0 & 0 & -Q_{xxzx} & Q_{xxyz} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 3Q_{g0}^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{xxyy} = -2Q_{du}^{(1)} + Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{xxzz} = Q_{du}^{(1)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{xxyz} = Q_{gb}^{(1)}$$

$$Q_{xxzx} = Q_{ga}^{(1)}$$

$$Q_{zzzz} = 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 8Q_{g0}^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{yzyz} = Q_{du}^{(2)} - 4Q_{g0}^{(1)} + Q_s^{(2)}$$

\* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxyy} \\ 0 & 0 & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & -Q_{xxyy} \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ -Q_{xxyz} & Q_{xxyz} & 0 & 0 & Q_{yzzx} & Q_{xxzx} \\ -Q_{xxzx} & Q_{xxzx} & 0 & -Q_{yzzx} & 0 & -Q_{xxyz} \\ -Q_{xxyy} & Q_{xxyy} & 0 & -Q_{xxzx} & Q_{xxyz} & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3Q_{du}^{(3)}$$

$$Q_{xxyz} = 2G_{f2}^{(1)}$$

$$Q_{xxzx} = -2G_{f1}^{(1)}$$

$$Q_{xxyy} = -2G_{faz}^{(1)} + 2G_{pz}^{(1)}$$

$$Q_{yzzx} = -4G_{faz}^{(1)} - G_{pz}^{(1)}$$

\* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = -2Q_{du}^{(4)} + Q_s^{(3)}$$

$$Q_{xyxy} = 4Q_{du}^{(4)} + Q_s^{(3)}$$

\* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & Q_{yzzx} & 0 \\ -Q_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzzx} = -G_{pz}^{(2)}$$

\* Rank 4 tensor (sa).

$$\begin{bmatrix} Q_{xxyz} & Q_{xxzx} & Q_{xxxxy} \\ -Q_{xxyz} & -Q_{xxzx} & Q_{xxxxy} \\ 0 & 0 & Q_{zzxy} \\ Q_{yzyz} & Q_{yzzx} & 0 \\ Q_{yzzx} & -Q_{yzyz} & 0 \\ Q_{xxzx} & -Q_{xxyz} & 0 \end{bmatrix}$$

$$Q_{xxyz} = G_{f2}^{(2)}$$

$$Q_{xxzx} = G_{f1}^{(2)}$$

$$Q_{xxxxy} = -G_{faz}^{(2)} + G_{pz}^{(3)}$$

$$Q_{zzxy} = 2G_{faz}^{(2)} + G_{pz}^{(3)} + 2G_{pz}^{(4)}$$

$$Q_{yzyz} = -3Q_{du}^{(5)}$$

$$Q_{yzzx} = -G_{faz}^{(2)} + G_{pz}^{(4)}$$

\* Rank 4 tensor (as).

$$\begin{bmatrix} Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & Q_{yzzx} & -Q_{yzyz} & -Q_{yzxx} \\ Q_{xyxx} & Q_{xyxx} & Q_{xyzx} & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzxx} = G_{f2}^{(3)}$$

$$Q_{yzyz} = -3Q_{du}^{(6)}$$

$$Q_{yzzx} = -G_{faz}^{(3)} + G_{pz}^{(6)}$$

$$Q_{yzxy} = G_{f1}^{(3)}$$

$$Q_{xyxx} = -G_{faz}^{(3)} + G_{pz}^{(5)}$$

$$Q_{xyzx} = 2G_{faz}^{(3)} + G_{pz}^{(5)} + 2G_{pz}^{(6)}$$

\* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxxy} & Q_{xxzy} & Q_{xxxz} & Q_{xxyx} \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & -Q_{xxxxy} & -Q_{xxzy} & -Q_{xxxz} & -Q_{xxyx} \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & Q_{zzxy} & 0 & 0 & -Q_{zzxy} \\ Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzxy} \\ -Q_{yzxy} & Q_{yzxy} & 0 & -Q_{yzxz} & Q_{yzzy} & Q_{yzxx} & -Q_{yzzx} & Q_{yzyz} & Q_{yzxx} \\ -\frac{Q_{xxyy}}{2} - \frac{Q_{xxxz}}{2} & \frac{Q_{xxyy}}{2} + \frac{Q_{xxxz}}{2} & 0 & -Q_{xxzx} & Q_{xxzy} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & -Q_{xxzx} & Q_{xxzy} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 3Q_{g0}^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{xxyy} = -2Q_{du}^{(1)} + Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{xxzz} = Q_{du}^{(1)} + 3Q_{du}^{(3)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{xxyz} = 2G_{f2}^{(1)} + G_{f2}^{(2)} + Q_{gb}^{(1)}$$

$$Q_{xxzx} = -2G_{f1}^{(1)} + G_{f1}^{(2)} + Q_{ga}^{(1)}$$

$$Q_{xxxy} = -2G_{faz}^{(1)} - G_{faz}^{(2)} + 2G_{pz}^{(1)} + G_{pz}^{(3)}$$

$$Q_{xxzy} = 2G_{f2}^{(1)} - G_{f2}^{(2)} + Q_{gb}^{(1)}$$

$$Q_{xxxz} = -2G_{f1}^{(1)} - G_{f1}^{(2)} + Q_{ga}^{(1)}$$

$$Q_{xxyx} = -2G_{faz}^{(1)} + G_{faz}^{(2)} + 2G_{pz}^{(1)} - G_{pz}^{(3)}$$

$$Q_{zzxx} = Q_{du}^{(1)} - 3Q_{du}^{(3)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{zzzz} = 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 8Q_{g0}^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{zzxy} = 2G_{faz}^{(2)} + G_{pz}^{(3)} + 2G_{pz}^{(4)}$$

$$Q_{yzxx} = -2G_{f2}^{(1)} + Q_{gb}^{(1)}$$

$$Q_{yzyz} = Q_{du}^{(2)} - 3Q_{du}^{(5)} - 4Q_{g0}^{(1)} + Q_s^{(2)}$$

$$Q_{yzzx} = -4G_{faz}^{(1)} - G_{faz}^{(2)} - G_{pz}^{(1)} + G_{pz}^{(4)}$$

$$Q_{yzxy} = -2G_{f1}^{(1)} - Q_{ga}^{(1)}$$

$$Q_{yzzz} = Q_{du}^{(2)} + 3Q_{du}^{(5)} - 4Q_{g0}^{(1)} + Q_s^{(2)}$$

$$Q_{yzxz} = -4G_{faz}^{(1)} + G_{faz}^{(2)} - G_{pz}^{(1)} - G_{pz}^{(4)}$$

\* Rank 4 tensor (a).

$$\begin{bmatrix} Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzz} & Q_{yzxz} & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & Q_{yzxz} & -Q_{yzzy} & -Q_{yzxx} & Q_{yzzx} & -Q_{yzyz} & -Q_{yzxx} \\ Q_{xyxx} & Q_{xyxx} & Q_{xyzx} & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$Q_{yzxx} = G_{f2}^{(3)}$$

$$Q_{yzyz} = -2Q_{du}^{(4)} - 3Q_{du}^{(6)} + Q_s^{(3)}$$

$$Q_{yzzx} = -G_{faz}^{(3)} - G_{pz}^{(2)} + G_{pz}^{(6)}$$

$$Q_{yzxy} = G_{f1}^{(3)}$$

$$Q_{yzzz} = 2Q_{du}^{(4)} - 3Q_{du}^{(6)} - Q_s^{(3)}$$

$$Q_{yzxz} = -G_{faz}^{(3)} + G_{pz}^{(2)} + G_{pz}^{(6)}$$

$$Q_{xyxx} = -G_{faz}^{(3)} + G_{pz}^{(5)}$$

$$Q_{xyzx} = 2G_{faz}^{(3)} + G_{pz}^{(5)} + 2G_{pz}^{(6)}$$

$$Q_{xyxy} = 4Q_{du}^{(4)} + Q_s^{(3)}$$

\* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & & & Q_{xxxx} \\ 0 & Q_{xxxx} & & & Q_{yyyy} \\ 0 & 0 & & & Q_{zzzz} \\ -Q_{xxxx} & Q_{yyyy} & & & Q_{yyyy} \\ Q_{zzxx} & 0 & & & 0 \\ 0 & Q_{xyxy} & & & -Q_{yyyy} \\ 0 & Q_{yzyz} & & & 0 \\ Q_{xxxz} & -Q_{yzyz} & -\frac{Q_{xxxx}}{2} + \frac{3Q_{xyyy}}{2} + Q_{yzyz} - Q_{yzzy} + Q_{zzxx} & & \\ \frac{Q_{xxxx}}{2} - \frac{Q_{xyyy}}{2} & 0 & & & -Q_{xxxx} \\ -Q_{yyyy} & -Q_{xxxx} & & & 0 \end{bmatrix}$$

$$Q_{xxxx} = -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 3Q_{g0}^{(1)} + 3Q_s^{(1)}$$

$$Q_{xxxz} = Q_{ga}^{(1)}$$

$$Q_{yyyy} = -Q_{gb}^{(1)}$$

$$Q_{zzzz} = 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 8Q_{g0}^{(1)} + 3Q_s^{(1)}$$

$$Q_{yyzz} = Q_{du}^{(1)} + 3Q_{du}^{(3)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{zzxx} = Q_{du}^{(1)} - 3Q_{du}^{(3)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{xxyy} = -2Q_{du}^{(1)} + Q_{g0}^{(1)} + Q_s^{(1)}$$

$$Q_{yzzz} = Q_{du}^{(2)} + 3Q_{du}^{(5)} - 4Q_{g0}^{(1)} + Q_s^{(1)}$$